

GCE AS MARKING SCHEME

WJEC / EDUQAS

AS ECONOMICS – COMPONENT 1

Practice Paper 3 (Hard)

About this marking scheme

This marking scheme follows the same band-based AO structure used in WJEC/Eduqas live mark schemes. For each question, marks are allocated against the relevant Assessment Objectives (AO1, AO2, AO3, AO4). Candidates may achieve different bands within a single question. Indicative content is illustrative — candidates do not need to include all of it to reach the highest band. Examiners should adopt a positive marking approach and credit any economically valid alternative.

Q1. Using calculations for at least two distinct price ranges along the demand curve, outline how PED changes along this demand curve and explain why a profit-maximising producer of an inferior necessity should be cautious about cutting prices at the bottom of the curve. [6]

AO1 2 marks	Good understanding – 2 marks Understands that PED becomes more inelastic moving down the curve (more elastic moving up). Limited understanding – 1 mark Understands the concept of price elasticity of demand.
AO2 4 marks	Excellent application – 4 marks Both elastic and inelastic calculations done correctly and expressed as a negative figure. Good application – 2–3 marks 3 marks: both inelastic and elastic calculations done correctly. 2 marks: one done correctly (or only one PED calculated for the whole curve). Limited application – 1 mark Some numbers identified but % change not used correctly.

Indicative content

AO1: PED measures the responsiveness of quantity demanded to a change in price. Along a straight, downward-sloping demand curve PED moves from elastic at high prices/low quantities, through unit-elastic at the midpoint, to inelastic at low prices/high quantities. Total revenue ($P \times Q$) is maximised at the unit-elastic midpoint.

AO2 — using the figure (P: 15 → 1.5; Q: 20 → 200):

Elastic region (top of curve), e.g. P £15 → £12 (–20%); Q 20 → 50 (+150%). $PED = +150 / -20 = -7.5$ (highly elastic).

Around the midpoint, e.g. P £8.25 → £6.75 (–18.2%); Q 105 → 125 (+19.0%). $PED \approx -1.04$ (approximately unit-elastic).

Inelastic region (bottom of curve), e.g. P £3 → £1.5 (–50%); Q 170 → 200 (+17.6%). $PED = +17.6 / -50 = -0.35$ (inelastic).

Implication for an inferior-necessity producer: at the bottom of the curve a further price cut reduces TR (because $|PED| < 1$, percentage rise in Q is smaller than percentage fall in P). Marginal cost of producing the extra units is still positive, so profit falls on two counts — revenue down and cost up. The producer should hold price near or above the unit-elastic point and avoid promotional discounts in the inelastic region.

Q2 (a). Using column 3 of Table 1, outline the differences between the three product categories using YED, and identify the category most exposed to a recession. [5]

Band	AO1 — 2 marks Is YED understood?	AO2 — 3 marks Is the data interpreted correctly?
3		3 – Excellent application Use of YED numbers to show understanding of both inelastic/elastic YED AND both normal/inferior goods.
2	2 – Good understanding of YED.	2 – Good application Use of YED for one matching pair plus one other element.
1	1 – Limited understanding of YED (may be by formula only).	1 – Limited application YED used to show either elastic/inelastic OR normal/inferior, but not both.
0	No valid.	No valid.

Indicative content

AO1: YED measures the responsiveness of demand to a change in income. Positive YED → normal good; negative YED → inferior good. $|YED| > 1$ = income-elastic; $|YED| < 1$ = income-inelastic. In a recession real incomes fall, so demand for income-elastic normal goods falls sharply, while demand for inferior goods can rise.

AO2 — for this paper:

Product 1 (luxury fashion, YED +4.8): strongly income-elastic, normal — a luxury good with a very pronounced response to changes in income.

Product 2 (discount supermarket basket, YED –2.8): income-elastic, inferior — a strong inferior good; demand rises sharply as incomes fall.

Product 3 (mid-tier household goods, YED +0.4): income-inelastic, normal — a necessity-type good with a mild positive response to income.

Most exposed to a recession: Product 1 — a 1% fall in income reduces demand by ~4.8%, by far the largest fall of the three categories.

Q2 (b). Using Table 1 and Figure 1, consider whether the change in German Real GDP between 2019 and 2021 will be harmful for the total revenue of the retailer, paying particular attention to the volatile pattern of GDP over the period. [6]

Band	AO2 (2) Answer in context?	AO3 (2) Reasoning on TR?	AO4 (2) Debated & judged?
2	Good application – both GDP data and YEDs used.	Good analysis – developed lines on impact of GDP on TR.	Good evaluation – reasoned judgement; counter-arguments developed.
1	Limited application – limited use of GDP/YEDs.	Limited analysis – superficial.	Limited evaluation – counterpoints underdeveloped, or one-sided judgement on sales proportions.
0	No valid.	No valid.	No valid.

Indicative content

AO2: German Real GDP shows a sharp V-shape: from around €830bn (Q1 2019) it fell to a trough near €750bn in Q2 2020 (pandemic shock), then recovered to ~€820bn by Q4 2021 — net incomes broadly back to pre-shock levels but with a deep mid-period trough. 55% of sales (Products 1 and 3) are normal goods; 35% of sales (Product 2) are strongly inferior; 30% (Product 1 alone) is the highly income-elastic luxury category. 10% of sales have unknown YED.

AO3: During the 2020 collapse: Product 1 demand falls sharply (very high +YED meets a large income fall) → significant revenue loss on the luxury line. Product 2 demand rises substantially (strong –YED meets falling income) → revenue gain on discount basket. Product 3 demand falls modestly (mild +YED). During the 2021 recovery the pattern reverses partially: Product 1 recovers strongly, Product 2 weakens. Total revenue = $\Sigma P \times Q$.

AO4: The V-shape matters: had GDP fallen and stayed low, Product 1 revenue losses would have been sustained and severe; had GDP risen smoothly, Product 2 would have lost ground steadily. The actual V-shape produces partial offsets within each sub-period. On balance, total revenue is likely to have been damaged in 2020 (Product 1 is highly income-elastic and its losses dominate) but largely recovered by end-2021 as luxury demand rebounded. Depends on: (i) the YED of the 10% unknown category — if it includes income-elastic premium lines, the swings are amplified; (ii) the retailer's ability to rebalance stock mix between sub-periods; (iii) German government furlough schemes which cushioned actual disposable incomes (so the income shock to households was smaller than the GDP shock suggests); (iv) shifts in e-commerce penetration during lockdowns. **Judgement:** short-run revenue harm in 2020 is highly likely; full-period (2019–2021) harm is moderate at most because the V-shape brings GDP back close to its starting point.

Q3 (a). Using a D&S; diagram, outline how the Argentinian central bank may previously have used capital controls and reserves to manage its exchange rate **above** the free-market equilibrium, identifying the shortage created at the managed rate. **[3]**

Band	AO1 (2) Diagram drawn accurately?	AO1 (1) Method outlined?
2	Good understanding. Appropriate D&S; diagram with the managed ER clearly above/below the free-market equilibrium, fully labelled.	
1	Limited understanding. Diagram attempted but with significant labelling errors or the free-market level not clearly marked.	1 mark. Correct outline of how the supply/demand of currency is used to manage the rate.
0	No valid.	No valid.

Indicative content

AO1 diagram: D&S; diagram for the Argentine Peso (ARS) against the US dollar. The managed ARS rate is drawn *above* the free-market equilibrium (i.e. the peso is held artificially strong against the dollar — fewer pesos per dollar than the market would produce). At the managed rate, quantity demanded of ARS exceeds quantity supplied, leaving a shortage of pesos / equivalently an excess demand for US dollars at the official rate. The shortage should be marked explicitly between Q_s and Q_d on the diagram.

AO1 method: The central bank used (i) capital controls: legal restrictions on the purchase of US dollars at the official rate (limits on individual purchases, restrictions on companies repatriating profits, prohibitions on dollar savings); (ii) selling US dollars from foreign-exchange reserves into the FX market to satisfy the official-rate demand for dollars and buying pesos back, which shifts the demand for ARS to the right; (iii) higher domestic interest rates to attract demand for ARS. Together these prevent the market from clearing at the lower free-market peso value.

Q3 (b). Using Figure 2 and the surrounding information, explain why the existence of the parallel "blue dollar" rate of \$1 : 1 000 ARS is strong evidence that the previous official rate was not the result of free-market forces. **[3]**

Band	AO2 (2) Data used to support claim?	AO3 (1) Reasoning explained?
2	Good application: precise reference to the trade data with understanding of net trade flows.	
1	Limited application: one piece of data used (imports OR exports, not both).	1 mark – Analysis: developed reasoning explaining the effect on the exchange rate.
0	No valid.	No valid.

Indicative content

AO2: Argentina runs a trade deficit with the US (\$8.7bn imports against \$5.4bn exports), generating structural excess demand for dollars in the market. The pre-devaluation official rate (\$1 : 366 ARS) is dramatically stronger than the parallel "blue dollar" rate (\$1 : 1 000 ARS) — a roughly 2.7-fold gap between the two rates for the same currency pair.

AO3: If the official rate genuinely reflected free-market clearing, no rational agent would pay 1 000 ARS for a dollar on the unofficial market when 366 ARS would buy the same dollar officially — the blue market would not exist. The persistence of a very large parallel-market premium is direct evidence that supply of dollars is rationed at the official rate (capital controls bind) and that the true market-clearing rate sits much closer to the blue dollar value. The official rate is therefore a managed rate held above the free-market equilibrium, not the equilibrium itself.

Q4. Using the data and economic analysis, discuss the strength of the relationship between Turkey's exchange rate and its economic growth, inflation and net exports between January 2021 and October 2022, considering how unconventional monetary policy (cutting rates while inflation rose sharply) may complicate the standard analysis. [9]

Band	AO1 (2)	AO2 (3)	AO3 (2)	AO4 (2)
3		Excellent application: all four data series used effectively. Answer thoroughly embedded in the data.		
2	Good understanding of ER and how it influences growth, inflation and net exports.	Good application: data used with clear reference to ER and at least 2 of the 3 outcomes.	Good analysis: developed lines on how variables link.	Good evaluation: reasoned judgement; counter-arguments developed.
1	Limited understanding (ER + one impact only).	Limited application (ER + at least one outcome).	Limited analysis – superficial.	Limited evaluation – counterpoints underdeveloped.
0	No valid.	No valid.	No valid.	No valid.

Indicative content

AO1: An ER depreciation raises import prices and lowers the foreign-currency price of exports — net exports may improve subject to the Marshall–Lerner condition and a J-curve lag. Import-price pass-through drives cost-push inflation. Standard policy response to high inflation is to *raise* interest rates, which both restrains AD and supports the currency by attracting capital inflows.

AO2: The Turkish Lira fell sharply against the US dollar — from roughly 1 USD = 7.5 TRY (Jan 2021) to 1 USD ≈ 18.6 TRY (Oct 2022), a depreciation of over 60%. Inflation rose from ~15% (Jan 2021) to a peak of ~85% (Oct 2022). Real GDP growth was volatile but remained positive on annual figures (around 7–11% YoY through much of the period). Net exports improved in nominal Lira terms but the trade deficit in USD terms widened due to surging energy import costs.

AO3: Standard theory predicts the large Lira depreciation should: (i) raise imported inflation substantially — clearly present (CPI rose 5–6x); (ii) improve net exports via cheaper Turkish exports — partially visible in export volumes but offset by expensive imports of energy and intermediate goods on which Turkish manufacturing depends; (iii) initially support GDP via export competitiveness but eventually drag on growth as real incomes are eroded by inflation. The relationship between ER and inflation appears very strong; the relationships with net exports and growth are weaker.

AO4: The strength of the ER–inflation link in Turkey is extreme but is amplified by unconventional policy: the central bank *cut* the policy rate (from 19% to 9% between 2021 and 2022) *while* inflation rose, weakening the Lira further and feeding back into more imported inflation — a destabilising loop that standard analysis would not predict. The ER–growth link is muddled: nominal growth remained high but real GDP per capita stagnated as inflation eroded purchasing power; consumption was front-loaded as households tried to escape from depreciating cash holdings. The ER–net-exports link is consistent with theory only in volume terms; the value of the trade deficit widened because Turkey is a net importer of energy and intermediate goods whose dollar prices spiked over the same period (Russia–Ukraine war).

Judgement: The relationship between ER and inflation is very strong (and made stronger by unorthodox monetary policy); the relationship with net exports is real but partially offset by external commodity shocks and the import-dependence of Turkish industry; the relationship with growth is largely indirect, operating through inflation's effect on real incomes. Standard textbook predictions hold qualitatively but are heavily distorted by the policy regime; the ER is best seen as one channel within a wider macroeconomic policy

experiment.

Q5. To what extent will the absence of enforceable private property rights over antibiotic effectiveness lead to market failure? Consider both consumption-side and production-side market failures. [8]

Band	AO1 (2) Market failure understood?	AO2 (2) In context?	AO3 (2) Link explained?	AO4 (2) Debated & judged?
2	Good – welfare loss / inefficient allocation defined precisely.	Good application: case used effectively throughout.	Good analysis: developed lines on lack of property rights → market failure.	Good evaluation: reasoned judgement; counter-arguments developed.
1	Limited – imprecise definition of market failure.	Limited application: case used superficially.	Limited analysis: superficial.	Limited evaluation: counterpoints underdeveloped.
0	No valid.	No valid.	No valid.	No valid.

Indicative content

AO1: Market failure is the inefficient allocation of resources, resulting in a welfare loss. In the presence of a negative externality $MSC > MPC$ and the free market over-produces or over-consumes; under a positive externality of production $MSB > MPB$ and the free market under-produces.

AO2: Antibiotic effectiveness behaves as a global common-access resource: it is rivalrous (each use depletes it) but non-excludable (no one can be prevented from using existing compounds once licensed). No country, firm or patient has an enforceable right to limit others' antibiotic use. Many economies allow over-the-counter sales; agricultural use is widespread. The pipeline of new antibiotics has slowed because commercial returns are limited.

AO3 — consumption-side market failure: Each prescriber and patient enjoys the full private benefit of treatment but bears only a tiny share of the marginal social cost of resistance, which is spread across all future users. $MPB \ll MSB\text{-net-of-resistance-cost}$, so consumption sits above the social optimum — a classic tragedy of the commons applied to a medical resource. Welfare loss accumulates as resistant strains emerge.

AO3 — production-side market failure: The development of a new antibiotic generates a positive externality — it restores resistance to society — but the developer cannot capture this social benefit. Worse, a successful new antibiotic is deliberately reserved (held back from prescribing) to preserve effectiveness, which destroys the commercial revenue that would have justified the R&D.; MPB to the developer is far below MSB to society. Result: under-investment in antibiotic R&D.; an empty pipeline, and a slow collapse of the resource.

AO4: The case for severe market failure: the externality is global, cumulative, and operates on a timescale (decades) longer than political cycles. Free-rider incentives are strong — every country gains if others restrict use while it does not. Both sides of the market fail simultaneously, an unusually deep market failure. **Counter — partial substitutes for property rights exist:** the WHO's GLASS resistance-surveillance scheme; national stewardship programmes restricting prescribing; "pull" incentives such as the UK's NHS subscription-style payment model for new antibiotics (which pays a developer a fixed sum regardless of volume, effectively buying back the externality); patent extensions and priority review vouchers; and EU restrictions on agricultural antibiotic use that approximate a global quota at regional scale. **However:** these mechanisms are unevenly applied, easily reversed, and cover a minority of global antibiotic use. The consumption externality is barely addressed in low-income economies. The production externality is only partially internalised by the most generous "pull" schemes. **Judgement:** the absence of enforceable property rights is the *fundamental* cause of severe market failure in both directions — over-consumption and under-production simultaneously. Domestic and regional policies mitigate but cannot substitute fully for global enforcement. The depth of market failure is therefore very high — among the strongest cases for coordinated intervention in modern microeconomics — and made more acute because the two failures reinforce each other.

Q6. Evaluate whether the proposed gambling package is likely to lead to higher prices, higher unemployment and a lower rate of economic growth. [9]

Band	AO2 (2) In context?	AO3 (3) Mechanisms explained?	AO4 (4) Debated & judged?
3		Excellent analysis: detailed lines for all three outcomes.	Excellent evaluation (4): reasoned judgement clearly set in context.
2	Good application: data/case used for at least two outcomes.	Good analysis: developed lines for two outcomes.	Good evaluation (2–3): reasoned judgement; counter-arguments developed.
1	Limited application: case used superficially or only one outcome.	Limited analysis: one outcome explained superficially.	Limited evaluation: counterpoints underdeveloped.
0	No valid.	No valid.	No valid.

Indicative content

AO2: The package: 1% statutory levy on gross gambling yield (ring-fenced for addiction treatment); TV and radio betting-advertising ban 5:30am–10:30pm; affordability checks above £150/month deposits; sports sponsorship ban from 2027; 24-hour cooling-off on stakes above £500 for online casino.

AO3 – Higher prices: The levy raises the marginal cost of every gambling transaction by 1% of gross yield — operators with normal margins will pass at least part of this through into worse odds and lower bonuses (the effective "price" of gambling). Affordability checks add compliance costs which shift supply left. Reduced promotional spending (advertising ban + sponsorship ban) removes promotional discounts customers currently enjoy, raising the effective price.

AO3 – Higher unemployment: Falling sales reduce derived demand for labour in betting shops, online operator customer-service teams, and sports-marketing agencies. The sports sponsorship ban (from 2027) directly removes a major revenue line for football, rugby and racing — knock-on job losses in those sectors. Compliance teams will grow but not enough to offset frontline losses. Regional concentration of betting shops in lower-income areas amplifies the local impact.

AO3 – Lower growth: Gambling contributes meaningfully to consumer spending (C); falling C reduces AD. Reduced expected industry profits depress I, especially digital R&D; spend. Sponsorship loss reduces sports-sector revenues. Offsetting: 1% levy funds NHS addiction services, raising productivity; reduced problem-gambling debt frees household income for productive consumption elsewhere.

AO4 – Higher prices: Likely on average, but odds and promotional spend are heterogeneous across product types — sports betting margins are tight and may absorb most of the levy, whereas online casino has scope to widen the house edge. The 24-hour cooling-off on £500+ stakes does not change unit prices but raises the time cost of high-stake casino play. **Unemployment:** Job losses in betting shops are likely but largely offset by substitution within consumer leisure spending (cinema, pubs, sports attendance — all labour-intensive sectors). Frictional rather than structural in most regions; structural in towns dependent on betting-shop high streets. **Growth:** Direct effect on GDP is small (UK gambling Gross Value Added is ~0.4% of GDP). The sponsorship pipeline disruption to professional sport is more material but rebrands quickly. Long-run growth effect through reduced gambling-related debt, suicide, family breakdown and lost productivity could be net positive — public-health groups' productivity claim has support in NHS England estimates. **Judgement:** Higher prices for some gambling products — yes, by design and incentive. Modest, sub-sectoral employment losses concentrated geographically. Headline growth effect is plausible but small in the short run; the long-run growth effect is ambiguous and could be net positive once productivity gains from reduced problem-gambling harms are scored. The Betting and Gaming Council's macro warnings appear overstated relative to the size of the sector; the public-health case is internally consistent.

Q7 (a). Using calculations in your answer, explain why the total wage cap in 2023 might have been said to be "too low". [4]

Band	AO2 (3) Are calculations used?	AO3 (1) Impact on wages explained?
3	Excellent application: total wage without cap compared to the cap using a 52-week or 365-day year.	
2	Good application: average wage per player comparison, OR total wage compared to cap using a 48-week year (12×4).	
1	Limited application: weekly total wage calculated only.	1 mark – Analysis: explanation of why the wage cap is too low.
0	No valid.	No valid.

Indicative content

AO2 (calculations):

Current implied annual wage bill at £6,250/week per player $\times$ 52 weeks $\times$ 24 players = £7,800,000 (£7.8m).

Or using 48 weeks ($12 \text{ months} \times 4$): $£6,250 \times 48 \times 24 = £7,200,000$ (£7.2m).

Maximum permitted average wage at the cap = $£4,800,000 \div 24 \text{ players} \div 52 \text{ weeks} = £3,846$ per week.

Gap between current total bill (52-week basis) and cap = $£7.8\text{m} - £4.8\text{m} = £3.0\text{m}$ per team — roughly 38% of the current total wage bill must be cut.

AO3: Because the cap (£4.8m) sits significantly below the implied current total annual wage bill (£7.8m), teams cannot continue paying current average wages to a full 24-player squad. Either average wages must fall by ~38%, or roughly a third of the squad must be released, or some combination of the two. For teams paying premium wages to attract international players above the league average, the cap bites harder still — making it implausible to retain top earners alongside a full senior squad. Hence the claim that the cap is "too low" to maintain competitive playing groups at present.

Q7 (b). Using a diagram in your answer, discuss whether the removal of the maximum price of labour was the **main** cause of the wage growth of elite American sports players since 1976. [7]

Band	AO1 (3) Diagram correct?	AO2 (2) In context?	AO4 (2) Debated & judged?
3	Excellent understanding: max-price diagram fully integrated; impact on wages on removal shown clearly.		
2	Good understanding: max-price diagram of impact on wages if removed; may include D for labour shift due to TV revenue.	Good application: uses data and inelastic PES of labour to support strong wage growth.	Good evaluation: reasoned judgement on whether abolition was the main cause; counter-arguments developed.
1	Limited: no diagram, but understanding of max-wage removal effect shown.	Limited application: limited use of data.	Limited evaluation: counterpoints underdeveloped.
0	No / incorrect diagram.	No valid.	No valid.

Indicative content

AO1 (diagram): Wage on y-axis, quantity of elite-player labour on x-axis. Steep (highly inelastic) S_L reflecting the scarcity of elite athletic talent; downward-sloping D_L . The pre-1976 reserve clause is drawn as a binding maximum-wage horizontal line below free-market equilibrium — excess demand at the cap is marked between Q_s and Q_d . On removal, wage rises to the new equilibrium W_{new} on the original D_L . A second outward shift $D_L \rightarrow D_{L2}$ reflects the ~60-fold real rise in broadcasting revenue plus growth in franchise valuations and international markets — the new equilibrium wage W_{new} is far above W_{old} . Because S_L is highly inelastic, almost all of the demand-side shift translates into higher wages rather than higher quantity.

AO2: Real average wages rose from \$51,500 (1976) to \$1.95m (2000) — a ~38-fold real rise — and to \$4.90m (2023) — a further ~2.5-fold real rise on top. The first leg (1976–2000) coincides directly with the introduction of free agency after the reserve clause was struck down. The second leg (2000–2023) coincides with the largest era of broadcasting-rights inflation in US sports history and the opening of global media markets. The supply of elite athletes is highly inelastic in the short and medium term because of the long lead-time and physical-talent constraints on development.

AO4: The removal of the reserve clause was a *necessary* condition for the wage explosion: without it, competitive bidding between teams was impossible and players were effectively monopsony-priced. The first leg of the wage rise (1976–2000) is therefore largely attributable to the policy change — bidding finally pushed wages toward the market-clearing level for inelastic supply. **However:** the further ~2.5-fold real rise from 2000 to 2023 cannot be attributed to the policy change (already in place for 24 years). This further rise is driven by demand-side shocks: the 60-fold real rise in league broadcasting revenue, the rise of cable/streaming-rights bidding wars, sponsorship globalisation, and franchise-valuation growth. Combined with inelastic PES, any rightward shift in D_L translates almost entirely into wages. Free-agent mobility (a knock-on of the policy change) also widened the bidding pool, amplifying the broadcasting-demand effect. **Judgement:** the removal of the maximum price was the *most important single trigger* for the wage explosion, and the dominant cause of the 1976–2000 rise. But the *main cause* of the absolute wage levels reached by 2023 is the interaction of two forces: the policy change unlocked the market, and the demand-side revenue boom drove wages to current levels against an inelastic supply curve. Either factor alone would have produced much lower wages today. The policy change opens the door; the broadcasting boom is what walks through it.